

Model Development Phase Template

Date	25 June 2026
Team ID	SWTID-2026-5981
Project Title	Credit Card Approval Prediction
Maximum Marks	4 Marks

Initial Model Training Code, Model Validation and Evaluation Report

The initial model training code is showcased through a screenshot. The model validation and evaluation report includes classification reports, accuracy, and confusion matrices for multiple models, presented through respective screenshots.

Initial Model Training Code:

Logistic Regression Model

```
def logistic_reg(xtrain, xtest, ytrain, ytest):  
    lr = LogisticRegression(solver='liblinear')  
    lr.fit(xtrain, ytrain)  
    ypred = lr.predict(xtest)  
    print("LogisticRegression")  
    print('Confusion matrix')  
    print(confusion_matrix(ytest, ypred))  
    print('Classification report')  
    print(classification_report(ytest, ypred))  
    return lr
```

Random Forest Classifier

```
def random_forest(xtrain, xtest, ytrain, ytest):  
    rf = RandomForestClassifier()  
    rf.fit(xtrain, ytrain)  
    ypred = rf.predict(xtest)  
    print("RandomForestClassifier")  
    print('Confusion matrix')  
    print(confusion_matrix(ytest, ypred))  
    print('Classification report')  
    print(classification_report(ytest, ypred))
```

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XG Boost Model



```
def g_boosting(xtrain, xtest, ytrain, ytest):  
    gb = GradientBoostingClassifier()  
    gb.fit(xtrain, ytrain)  
    ypred = gb.predict(xtest)  
    print("GradientBoostingClassifier")  
    print('Confusion matrix')  
    print(confusion_matrix(ytest, ypred))  
    print('Classification report')  
    print(classification_report(ytest, ypred))
```

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Decision Tree Model

```
def d_tree(xtrain, xtest, ytrain, ytest):
    dt = DecisionTreeClassifier()
    dt.fit(xtrain, ytrain)
    ypred = dt.predict(xtest)
    print("DecisionTreeClassifier")
    print('Confusion matrix')
    print(confusion_matrix(ytest, ypred))
    print('Classification report')
    print(classification_report(ytest, ypred))
```

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Model Validation and Evaluation Report:

Model	Classification Report	Accuracy	Confusion Matrix
Logistic Regression	<pre>Classification report precision recall f1-score support 0 1.00 0.99 0.99 1037 1 0.99 0.99 0.99 885 accuracy: 0.99 0.99 0.99 1942 macro avg: 0.99 0.99 0.99 1942 weighted avg: 0.99 0.99 0.99 1942</pre>	99%	<pre>LogisticRegression Confusion matrix [[1025 12] [5 900]]</pre>
Random Forest Classifier	<pre>Classification report precision recall f1-score support 0 0.99 0.99 0.99 1037 1 0.99 0.99 0.99 885 accuracy: 0.99 0.99 0.99 1942 macro avg: 0.99 0.99 0.99 1942 weighted avg: 0.99 0.99 0.99 1942</pre>	98%	<pre>RandomForestClassifier Confusion matrix [[1022 15] [8 897]]</pre>

Gradient Boosting Classifier	<pre> Classification report precision recall f1-score support 0 1.00 1.00 1.00 1037 1 1.00 1.00 1.00 983 accuracy 1.00 1.00 1.00 2020 avg prc 1.00 1.00 1.00 2020 weighted avg 1.00 1.00 1.00 2020 </pre>	100%	<pre> GradientBoostingClassifier Confusion matrix [[1036 1] [4 981]] </pre>
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Decision Tree Classifier	<pre> Classification report precision recall f1-score support 0 1.00 1.00 1.00 1037 1 1.00 1.00 1.00 983 accuracy 1.00 1.00 1.00 2020 avg prc 1.00 1.00 1.00 2020 weighted avg 1.00 1.00 1.00 2020 </pre>	100%	<pre> DecisionTreeClassifier Confusion matrix [[1035 2] [1 984]] </pre>
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